Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice Thematic Assembly Centre
Arctic Ocean - High Resolution Sea Ice Information
L3 and L4

SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023
SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

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1.1	May 2025	II, V	Added information about new upstream data from Sentinel-1 and Radarsat Constellation Mission.	M. B. Kreiner, T. Wulf,	C. Wettre

ACRONYM TABLE

AI	Artificial Intelligence
AMSR2	Advanced Scanning Microwave Radiometer 2
ASIC	Automatic Sea Ice Concentration
ASIP	Automated Sea Ice Products
AVHRR	Advanced Very High Resolution Radiometer
CDR	Climate Data Record
CMEMS	Copernicus Marine Environment Monitoring Service
CF	Climate Forecast (convention for NetCDF)
DL	Deep Learning
DMI	Danish Meteorological Institute
(I)CDR	(Interim) Climate Data Record
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EW	Extra-Wide Swath
FLZ	Floe Size
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
FTP	Protocol to download files
GLO	Global
GRDM	Ground Range Detected Multilooked
HH	Horizontal transmit, Horizontal receive SAR polarization
HV	Horizontal transmit, Vertical receive SAR polarization
ICDR	Interim Climate Data Record
L3	Level-3
L4	Level-4
MDS	Marine Data Store
MET	Norwegian Meteorological Institute
MFC	Monitoring and Forecasting Centre
mIoU	Mean Intersection over Union
MSIC	Manually derived ice chart
MY	Multi-year

MYNRT	Mylti-year and near-real-time
NERSC	Nansen Environmental Remote Sensing Centre
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OSI SAF	Ocean & Sea Ice Satellite Application Facility
PUM	Product User Manual
QUID	Quality Information Document
RCM	Radarsat Constellation Mission
RMSD	Root Mean Square Difference
SAR	Synthetic Aperture Radar
SI	Sea Ice
SIC	Sea Ice Concentration
SSMIS	Special Sensor Microwave Imager / Sounder
SOD	Stage-of-development
SST	Sea Surface Temperature
TAC	Thematic Assembly Center
VIIRS	Visible Infrared Imaging Radiometer Suite
WMO	World Meteorological Organization

KEY DEFINITIONS

Glacier ice	Ice that has formed by the accumulation and compression of snow over time.
Marginal ice zone	The region of an ice cover which is affected by waves and swell penetrating into the ice from the open ocean". It is a transitional zone between open water and dense pack ice.
Multi-year ice	Sea ice that has survived at least one summer melt season
Pack ice	Sea ice that is not attached to the shoreline and drifts freely on the ocean, driven by winds and currents.
Sea Ice Concentration	Fraction (percentage) of the sea surface area of the grid cell containing sea-ice.
Sea Ice floe size	An ice floe is a relatively flat piece of floating ice that is at least 20 meters in width. Sea ice floes are classified by size, from the smallest to the largest: ice cakes (less than 20 meters), small floes (20–100 m), medium floes (100–500 m), big floes (500–2,000 m), vast floes (2–10 km), and giant floes (over 10 km).
Stage-of-development	Sea ice stage-of-development is classified by its age and thickness, ranging from young ice to multi-year ice.
Thick first-year ice	Sea ice thicker than 120 cm that has not survived at least one summer melt season.
Thin first-year ice	First-year sea ice with a thickness between 30 and 70 centimeters.
Vast & giant floes	A vast sea ice floe is 2 to 10 kilometers across, while a giant floe is greater than 10 kilometers across.
Young Ice	A transitional stage of sea ice with a thickness of 10 to 30 centimeters.
Level-2 (L2) data product	Geophysical variables derived from Level 1 source data on the Level 1 grid (typically the satellite swath projection).
Level-3 (L3) data product	L2 data granules remapped to a regular latitude/longitude grid. In this project, a typical L3 file may contain all the observations from one or more instruments in a 24-hour period.
Level 4 (L4) data product	Observations from multiple instruments using an analysis system (e.g., optimal interpolation) to produce a gridded, gap-free product.

RELEVANT DOCUMENT LIST

CMEMS-SEAICE-SQO-011-002-023-024	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated products pages[12] under Documentation
CMEMS-SEAICE-PUM-011-023-024	The Product User Manual presents the product details along with file structure. It can be found in the dedicated products pages under Documentation.

[1] See https://data.marine.copernicus.eu/product/SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023/description
[2] See https://data.marine.copernicus.eu/product/SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024/description

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This quality information document (QUID) covers the *Arctic Ocean - High Resolution Sea Ice Information L3 and L4* products (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 and SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_024) disseminated as part of the Copernicus Marine Environment Monitoring Service (CMEMS) Sea Ice Thematic Assembly Centre (SI TAC) since November 2024. The level-3 (L3) and level-4 (L4) products are both Multi-year and Near-real-time 'MYNRT' and contain a multiyear (MY, reprocessed) and a near real time (NRT) dataset each. The products are produced by the Danish Meteorological Institute's Automated Sea Ice Products (DMI-ASIP) production system. Figure 1 provides a schematic overview of the collection of the two products and four datasets.

The L3 daily product provides sea ice concentration, stage-of-development and floe size information with associated uncertainties, delivered as a NRT dataset and a corresponding MY dataset extending the product back in time, covering the ten years of the Sentinel-1 mission from October 2014 to October 2024. The L3 product's MY and NRT datasets are delivered on a 0.5 km spatial grid.

The corresponding L4 product is based on the L3 product's sea ice concentration information and associated uncertainty, resampled to a 1.0 km spatial grid and 'gap-filled' with European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) Ocean & Sea Ice Satellite Application Facility (OSI SAF) microwave radiometer-based sea ice concentration products: the L4 MY dataset uses the OSI-458 climate data record (CDR) and L4 NRT dataset uses the OSI-408 NRT sea ice concentration product. All four datasets are delivered as NetCDF format with a polar stereographic projection.

The quality assessment reported in this QUID has been conducted for the L3 product MY dataset full period from October 3rd 2014 to October 2nd 2024. The L3 NRT dataset that extend the MY dataset from October 3rd 2024 and onwards is produced using the same production chain (implemented in operational mode, not in reprocessing mode). The L4 product, being a resampled version of the L3 product, as well as the OSI SAF sea ice information in the L4 product, is not included in the quality assessment.

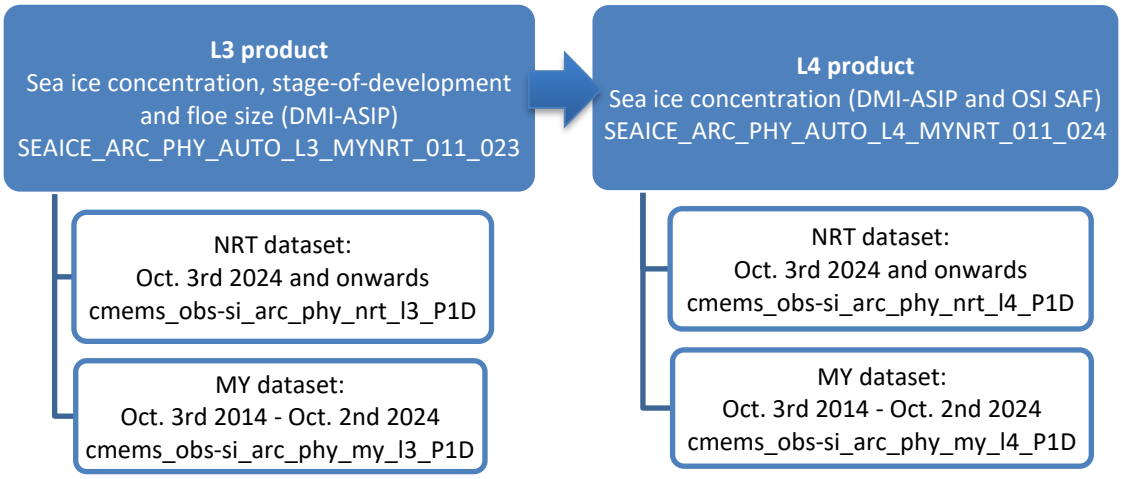

Figure 1 - Schematic overview of the L3 and L4 MYNRT (Multi-Year and Near-Real-Time) products, their inherent NRT and MY datasets and the time series they cover.

I.2 Summary of the results

The quality of the *Arctic Ocean - High Resolution Sea Ice Information L3* product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) is assessed in comparison with two different ice charts datasets from The Danish Meteorological Institute (DMI) Greenland Ice Service[3] and the National Ice Center (NIC)[4], respectively, as well as against the OSI SAF microwave radiometer-based sea ice concentration product. The product variables assessed are sea ice concentration, stage-of-development and floe size. Results for the three sea ice variables are summarised below.

Sea Ice Concentration (SIC): The assessments against the DMI and NIC ice chart datasets give inter-annual average Pearson's correlation coefficients of 0.93 ± 0.07 and 0.92 ± 0.03, respectively, suggesting very high correlation. In both assessments the seasonal results show highest correlations with very low standard deviations in months from June to September, and slightly lower correlation with somewhat larger standard deviations from October to May. This can be explained to a large degree by ice charting methodology and the use of SIC intervals. The assessment against the OSI SAF SIC product give an inter-annual average Pearson's correlation coefficients of 0.85 ± 0.02, suggesting high correlation, though not as good as for the ice charts. The high correlation applies for all months, only the sea ice minimum-month of September and October show slightly lower correlation. The largest differences between the two datasets are seen in the marginal ice zone, where the L3 product systematically provide higher SIC values than the OSI SAF product. This is most pronounced in the summer months (summer half-year is here defined as April to September) and can be explained by the coarser resolution OSI SAF product not being able to resolve the ice cover at intermediate and low SICs, nor the presence of melt-ponds on the ice.

Sea ice stage-of-development (SoD): The assessment of the products five SoD classes against the DMI and NIC ice chart datasets show best correspondence for the SoD classes: thick first-year ice (FYI) and multi-year ice (MYI) with inter-annual average F1 scores of 0.60 and 0.80, respectively, and a weak correlation for the two thinnest SoD classes: young Ice (YI) and thin first-year ice, and even less for the glacier ice class. This could partly be explained by the YI and thin FYI classes only being represented in

[3] See https://www.dmi.dk/gronland/is (last accessed 01/11/2025. Only in Danish).
[4] See https://usicecenter.gov/ (last accessed 01/11/2025).

the datasets in part of the year, and the glacier ice class only represented in minor areas along the Greenland coast. The reasonably good agreement applies both in summer and winter, with the L3 product having a tendency towards providing an older/thicker SoD class than the ice chart. The overall SoD class inter-annual average F1 score of 0.42 ± 0.07 and 0.45 ± 0.10, for the NIC and DMI dataset assessments respectively, which indicates a poor performance.

Sea Ice floe size (Flz): The assessment of comparing the products four Flz classes against the DMI ice chart dataset show best correspondence for the Flz class vast & giant floes, that is widely used in the L3 product and by the ice chart for the pack ice region in Central Arctic, Northeast and Northwest Greenland ice areas. There is a weaker correlation for the three smaller Flz classes, at least when compared to the well-represented vast & giant floe class. Since those three smaller floe size categories are primarily found in the more dynamical marginal ice zone, any time difference between the ice chart and L3 product can result in Flz class mismatch on the grid level and this may to some degree explain the low performance for those three classes. The reasonably good agreement applies both in summer and winter (winter half-year is here defined as October to March), for the minor overlapping sea ice area that is covered. Despite the good and reliable performance for the vast & giant floe class, the weak performance of three smaller Flz classes results in an overall inter-annual average F1 score of 0.39 ± 0.08, which indicate a poor performance.

I.3 Estimated Accuracy Numbers

In this section we provide a description and a summary of the metrics used for the comparison of the product with NIC ice charts, DMI ice charts and OSI SAF sea ice concentration product (OSI-SAF). The accuracy units and metrics for the products three variables are shown in the Table 1 below.

Table 1 – Product variables and units, accuracy metrics description and numbers from the different comparisons with NIC ice charts (NIC), DMI ice charts (DMI) and the OSI SAF sea ice concentration product (OSI SAF).

Variable	***Units***	***Metric***	***Accuracy numbers, mean ± std dev.***
Sea ice concentration (SIC)	-	Pearson's correlation coefficient, *r*: measures linear correlation between the product SIC and the reference datasets SIC. Goes from -1 to 1, where 1 represent a perfect correlation.	DMI: 0.93 ± 0.07
			NIC: 0.92 ± 0.03
			OSI SAF: 0.85 ± 0.02
Sea ice stage-of-development (SoD)	1 (class index)	F1 score: indicates the reliability of the SoD classification by the model, described as the harmonic mean of the *precision* and *recall*. Ranges from 0 to 1, where the higher the value, the higher the model reliability.	DMI: 0.45 ± 0.10
			NIC: 0.42 ± 0.07
Sea ice floe size (Flz)	1 (class index)	F1 score: indicates the reliability of the Flz classification by the model.	DMI: 0.39 ± 0.08

II PRODUCTION SYSTEM DESCRIPTION

The *Arctic Ocean - High Resolution Sea Ice Information L3 and L4* products are produced operationally by the Production Unit at the Danish Meteorological Institute (DMI) and provided to CMEMS through the SI TAC Production Centre at the Norwegian Meteorological Institute. See details in Table 2.

Table 2 - Product names, description and system version, SI TAC production unit and production centre.

Products	***Product description***	***Version***	***Production unit (PU)***	***Production Centre (PC)***
SEAICE_ARC_PHY_AUTO _L3_MYNRT_011_023	Arctic Ocean - High Resolution Sea Ice Information L3	2.0	SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO
SEAICE_ARC_PHY_AUTO _L4_MYNRT_011_024	Arctic Ocean - High Resolution Sea Ice Information L4			

The core of the production system is the DMI-ASIP deep learning model version 2.0, that takes as input synthetic aperture radar (SAR) imagery from Sentinel-1 (since October 2014) and Radarsat Constellation Mission (RCM, since November 2025) and AMSR2 microwave radiometer data. The DMI-ASIP model outputs sea ice information – sea ice concentration, stage-of-development and floe sizes - with associated uncertainty information (The DMI-ASIP model version 1.0 did only output sea ice concentration information). In the level-2 production this sea ice information is derived per individual Sentinel-1 and RCM image on an 80 m grid.

provides a schematic overview of the production system. The level-3 (L3) production takes in all level-2 (L2) products generated for individual Sentinel-1 and RCM images acquired within a day (UTC time) and collects them into a daily mosaic with the newest data on top, on a 0.5 km polar-stereographic grid. The daily L3 products contain the three variables, sea ice concentration, stage-of-development, floe sizes and the variables measuring the associated uncertainty information. In parallel to the level-3 production, the level-4 (L4) production takes the level-2 products generated for individual Sentinel-1 and RCM images acquired within a day (UTC time) and collects them into a daily mosaic on a 1.0 km polar-stereographic grid. Only the L2 sea ice concentration information and associated uncertainty information is used for the L4 production. Grid cells with no sea ice information from the L2 products are gap-filled with a daily EUMETSAT OSI SAF sea ice concentration product that is resampled onto the 1.0 km grid.

Two individual production systems have been established for the generation of the MY and NRT datasets, respectively, based on differences in input data and operational criteria. However, the two systems are alike in terms of processing steps from L2 to L3 and L4. A detailed description of the production system can be found in the Product user manual (PUM, 2024).

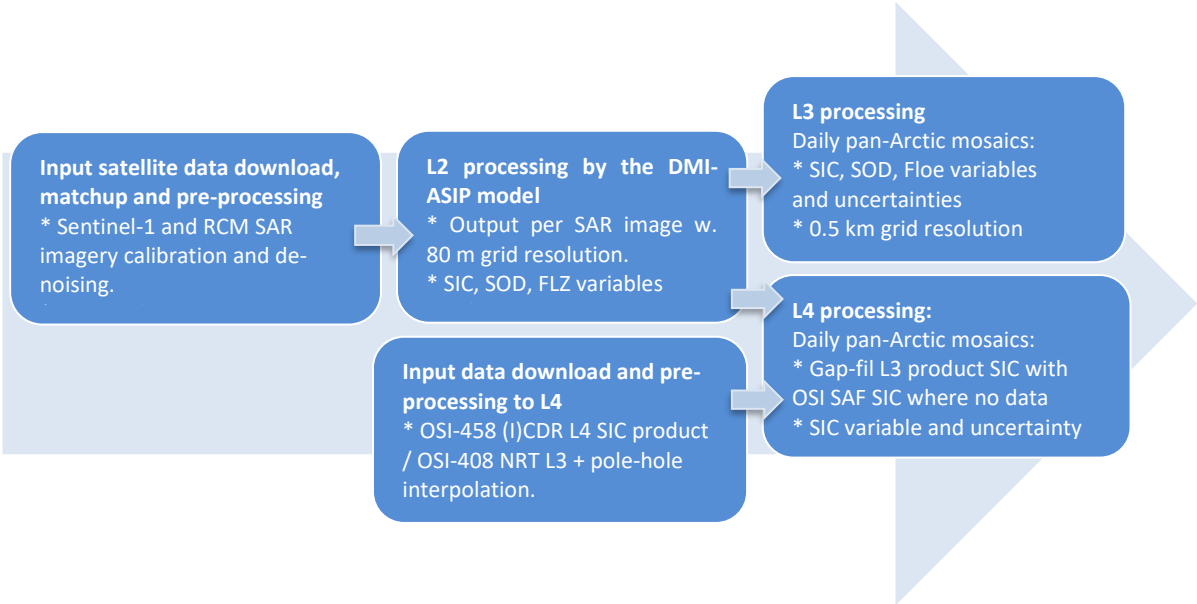

Figure 2 – Flow chart of the production system, from download of input satellite data and auxiliary OSI SAF data products, to the production of level-2 and hence the final level-3 and level-4 products for CMEMS.

III VALIDATION FRAMEWORK

This section describe the validation framework for each of the products three sea ice variables; sea ice concentration, stage-of-development and floe size, in the following three subsections, respectively.

The quality of the *Arctic Ocean - High Resolution Sea Ice Information L3* product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) is assessed in comparison with three different reference datasets:

- Greenland regional ice charts produced by the DMI Greenland Ice Service,
- Arctic ice charts produced by the National Ice Center (NIC), and
- EUMETSAT OSI SAF OSI-408-a sea ice concentration product.

See Table 3 for an overview of the reference datasets used for the assessments of each of the products´ three sea ice variables: sea ice concentration, stage-of-development and floe size. Examples of reference dataset files are shown in , and .

Table 3 – Overview of the ice chart reference datasets used for the quality assessment.

Reference dataset	***Variables***	***Area domain***	***Data source***
National Ice Center (NIC) ice charts	• Sea ice concentration (SIC) • Stage-of-development (SoD)	Pan-Arctic. See .	Shapefiles produced once a week, from Oct. 7th 2014 to Oct. 4th 2024. Available at: https://usicecenter.gov/Products/
Danish Meteorological Institute (DMI) ice charts	• Sea ice concentration (SIC) • Stage-of-development (SoD) • Floe size (Flz)	Greenland waters. See .	Shapefiles, produced every day, from Jan. 1st 2022 to Oct. 2nd 2024. Shapefiles available from DMI upon request. Redistributed as product 011_002.
EUMETSAT OSI SAF OSI-408-a	• Sea ice concentration (SIC)	Pan-Arctic. See .	NetCDF files, produced every day, from Jan. 1st to Dec. 31st 2021. Available from: https://thredds.met.no/thredds/osisaf/osisaf.html

The ice charts are a detailed interpretation of primarily SAR satellite imagery and a subsequent mapping procedure is carried out by skilled (experienced and trained) ice analysts drawing polygons of fairly homogeneous sea ice areas. National Ice Services classify the ice variables, such as concentration, stage-of-development and floe sizes, according to the World Meteorological Organization (WMO) SIGRID-3 standard (WMO, 2010), though some differences in the ice charting methodology occur between ice services[5]. The different methodologies may introduce biases to the ice chart datasets, and also it is to be assumed that the manual charting performed by different ice analysts introduces a bias to the individual ice charts (QUID, 2023).

The DMI regional ice charts are produced with irregular frequency (up to several times a day, depending on the acquisition of relevant SAR imagery) and are primarily used for strategic and tactical planning

[5] For a discussion of these, please see https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-002-012.pdf

within the offshore and shipping community. The DMI ice charts contain information about the sea ice concentration (in 10 % increments i.e., 0%, 10%, 20%, etc.), stage-of-development and floe sizes.

The NIC ice charts cover all of the Arctic and are produced once a week (but production frequency has varied over time (i.e., twice a week from Oct. 2013 – Jan. 2015, every second week from Dec. 2007 – Oct. 2013, and again weekly from Feb. 2006 – Dec. 2007), based on all available and useful satellite data. The NIC ice charts contain information about the sea ice concentration (in 20% intervals: 10-30%, 20-40%, etc.) and stage-of-development.

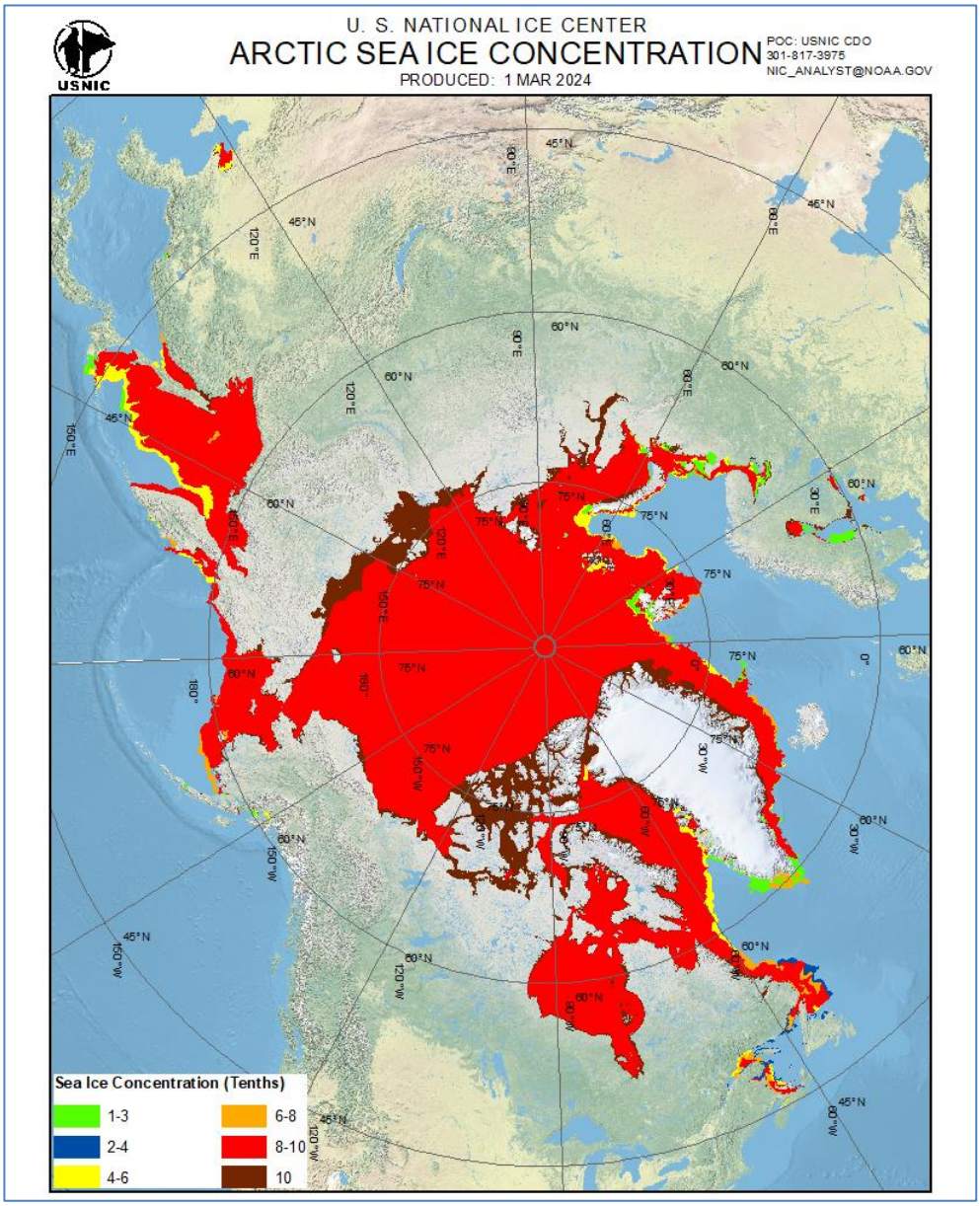

Figure 3 – Example of a National Ice Center (NIC) weekly Arctic ice chart of ice concentration, produced for March 1st 2024. A corresponding ice chart of sea ice stage-of-development is produced for the same date (not shown here) and the information is delivered with the underlying shapefile.

The two ice chart datasets have different properties in terms of sea ice parameters (SIC, SoD and FLZ versus SIC and SoD only), update frequency (two times weekly versus weekly) and spatial detail (regional versus pan-Arctic), that make them both relevant for the product quality assessment. The NIC ice charts are available for the whole product period from 2014 to 2024. The DMI ice charts, however, can only be used in a quality assessment of the product from 2022 to 2024, since the DMI ice charts that were produced and available from before that (from 2018 to 2021) have been used for training of the DMI-ASIP model.

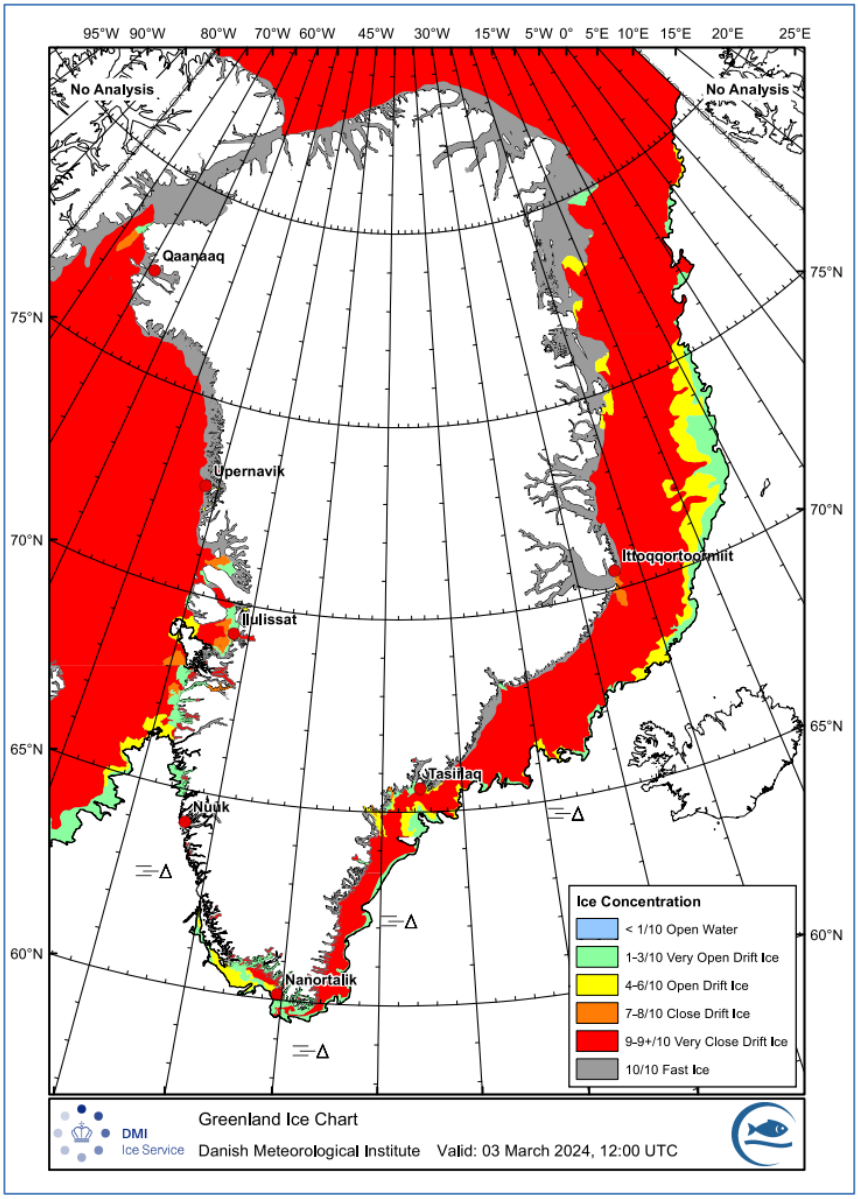

Figure 4 - Example of a DMI Greenland ice chart of ice concentration, produced for March 3rd 2024. Corresponding sea ice stage-of-development and floe size information is also produced and delivered with the underlying shapefile.

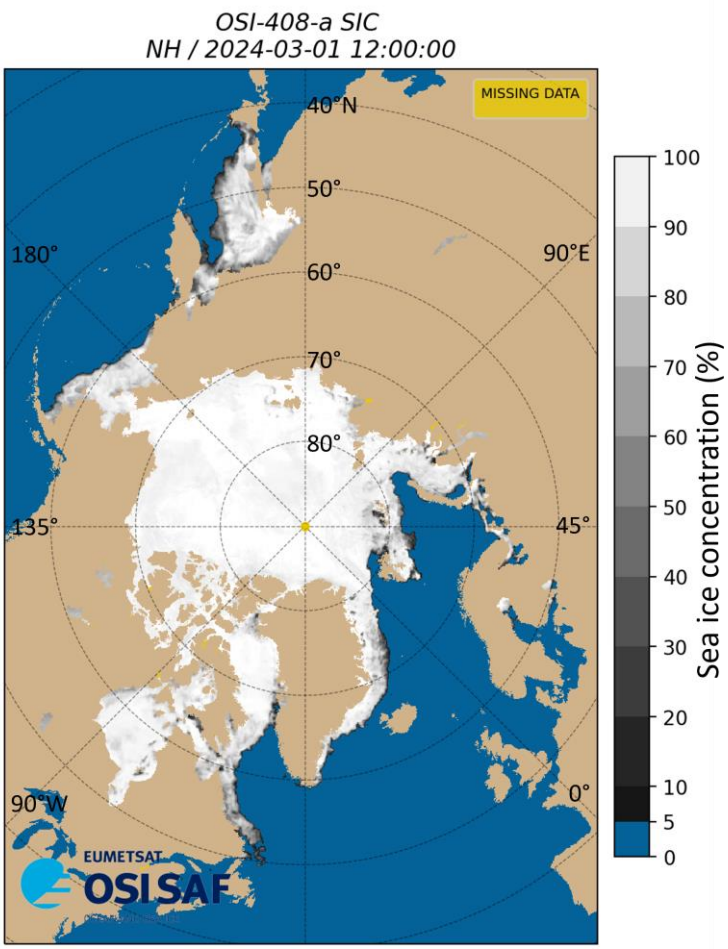

Figure 5 - Example of a EUMETSAT OSI SAF OSI-408-a sea ice concentration product, produced for March 1st 2024.

The third reference dataset used in the assessment is the EUMETSAT OSI SAF OSI-408-a sea ice concentration dataset covering the Arctic (except for a small area at the North Pole), automatically derived from AMSR2 satellite microwave radiometer data and provided daily at 25 km grid resolution. The OSI SAF dataset is included in the quality assessment of the L3 product, to assess the relative difference of the high resolution SAR based SIC product to the coarse resolution microwave radiometer based OSI SAF SIC product.

III.1 Sea Ice Concentration

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product sea ice concentration (SIC) variable is compared to the NIC ice charts, DMI ice charts and OSI SAF product. Both the DMI charts and OSI SAF product are produced daily, and validations are carried out day to day. NIC charts are produced weekly, and validation is performed by averaging the automatic ice product to match the chart.

The reference datasets are resampled to the products 0.5 km polar-stereographic grid (at the same time, the NIC ice concentration intervals are averaged: e.g., 10-30% to 20%) and the difference in sea ice concentration between the product and reference dataset is calculated for every grid cell.

All product grid values are 'binned' for each unique value of the reference data. Binned values are then averaged and metrics computed for the averaged values. For sea ice concentration the Pearson's

correlation coefficient and standard deviation is calculated for the annual average values as a measure of the linear correlation between the two datasets. Results are shown in Figures and Tables in Chapter IV.

III.2 Sea Ice Stage-of-Development

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product stage-of-development (SoD) variable is compared to the NIC ice charts and DMI ice charts. The product has five SoD categories:

- young ice,
- thin first-year ice (thin FYI),
- thick first-year ice (thick FYI),
- multiyear (MY) ice,
- glacier ice.

These categories are compared to the dominant ice type classes in NIC ice charts and DMI ice charts. Ice chart grid cells without a dominant ice type class are not considered in the assessment. The difference in sea ice SoD class between the product and reference dataset is calculated for every grid cell. The difference is set to 0 if both product and reference dataset are in the same class (i.e. if both the product and reference dataset show multiyear ice), and the difference can range from -3 to 3, where 3 indicates the product having an older/thicker SoD class than the reference dataset, 0 represents perfect agreement, and -3 signifies the product having a thinner SoD class than the reference dataset.

Annual average statistics are derived as described for the SIC assessment in Section III.1.

A confusion matrix is computed where an element of the matrix at the row *r* and column *c* is the number of samples predicted in the class *r* over the number of samples in the class *c*. A perfect classification would lead to a diagonal confusion matrix with 1 on the diagonal and 0 elsewhere. Each column sums up to 1. The accuracy per class (diagonal of confusion matrix) is the ratio of the correctly predicted samples of a given class over the total number of samples of the same class.

A *precision* metric is computed for each class. This metric indicates how often the model correctly predicts each class and it is calculated by dividing the number of correct positive predictions by the total number of predicted instances (both true and false positives). Likewise, the *recall* is calculated. The metric measures how often the model correctly predicts positive instances (true positives) from all the actual positive instances (number of true positives divided by the number of positive instances). Finally, the F1-score is calculated. It indicates the reliability of a model, described as the harmonic mean of the precision and recall of the classification model.

III.3 Sea Ice Floe Size

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product floe size (Flz) variable is compared to DMI ice charts.

The product has four Flz categories;

- small floes,
- medium floes,
- big floes,
- vast & giant floes

that are compared to the dominant floe size classes of the DMI ice charts. Ice chart grid cells without a dominant floe size class are not considered in the assessment.

The difference in sea ice Flz class between the product and reference dataset is calculated for every grid cell; the difference is set to 0 if both product and reference dataset are in the same class, and it is set to 1/-1 if the difference is in the order of a neighbouring Flz class, etc. The difference can range from -3 to 3, where 3 indicates the product having a larger Flz class than the reference dataset, 0 represents perfect agreement, and -3 signifies the product having a smaller Flz class than the reference dataset.

Annual average statistics are derived as described for the SIC assessment in Section III.1. Similar statistics are derived from the Flz assessment, as is done for the SoD assessment described above. A confusion matrix and F1 scores are computed.

IV VALIDATION RESULTS

Validation results are presented for each of the *Arctic Ocean - High Resolution Sea Ice Information L3* products three sea ice variables: sea ice concentration, stage-of-development and floe size in the following sections, respectively.

IV.1 Sea Ice Concentration

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product sea ice concentration (SIC) variable is compared to the NIC ice charts, DMI ice charts and OSI SAF product. The assessment results in the form of SIC difference plots and Pearson´s coefficient statistics are presented and discussed in the following sections.

IV.1.1 Comparison with NIC ice charts

The L3 product SIC values have been compared to the NIC ice chart SIC values (the mean SIC derived from the NIC SIC intervals) for all timely matches of the two datasets throughout the years. The SIC linear correlation is assessed by calculating the Pearson's coefficient. Figure 7 shows an overview of the assessment results in a time series plot of the annual average Pearson correlation coefficient for the sea ice concentration (SIC) comparison of NIC ice charts with the L3 product from 2014 to 2024. The same numbers (i.e., annual average Pearson coefficient and standard deviation), are provided in Table 4. Monthly average correlation coefficients ranges from 0.86 to 0.97, suggesting a very high correlation, with an interannual average and standard deviation of 0.93 ± 0.07.

Figure 11 (in chapter IV 1.3) zooms in on the seasonality of the correlation. This Figure shows a time series plot of the monthly average Pearsons coefficient for the sea ice concentration (SIC) comparison of NIC ice charts (and OSI SAF product and DMI ice charts) with the L3 product in the one year of 2021. In the year 2021 the L3 product has better spatial coverage than other years of the dataset, benefitting from the full operation of both the Sentinel-1A and -1B satellite SAR sensors. Monthly average correlation coefficients ranges from 0.86 to 0.97, suggesting a high correlation. Highest correlations with lower standard deviations are seen in the summer from June to September, whereas the correlation from October to May is slightly less good and with somewhat larger standard deviations. This is to a large degree caused by the use of SIC intervals in the ice charts, where the pack ice in the Central Arctic Ocean usually has one big polygon of 60-80% ice in summer and 80-100% ice in winter (see the NIC ice chart example in). The L3 product will usually provide a SIC of around 100% ice in that area in winter and >70% ice in summer. When compared against the average value of the ice chart SIC interval, it will cause systematic positive differences about 10-20% in this large area. This is evident from the example plots of the L3 product SIC, NIC ice chart SIC and the SIC difference shown in Figure 6, where the top panel is in winter (March 4th 2021) and bottom panel is in summer (September 16th 2021).

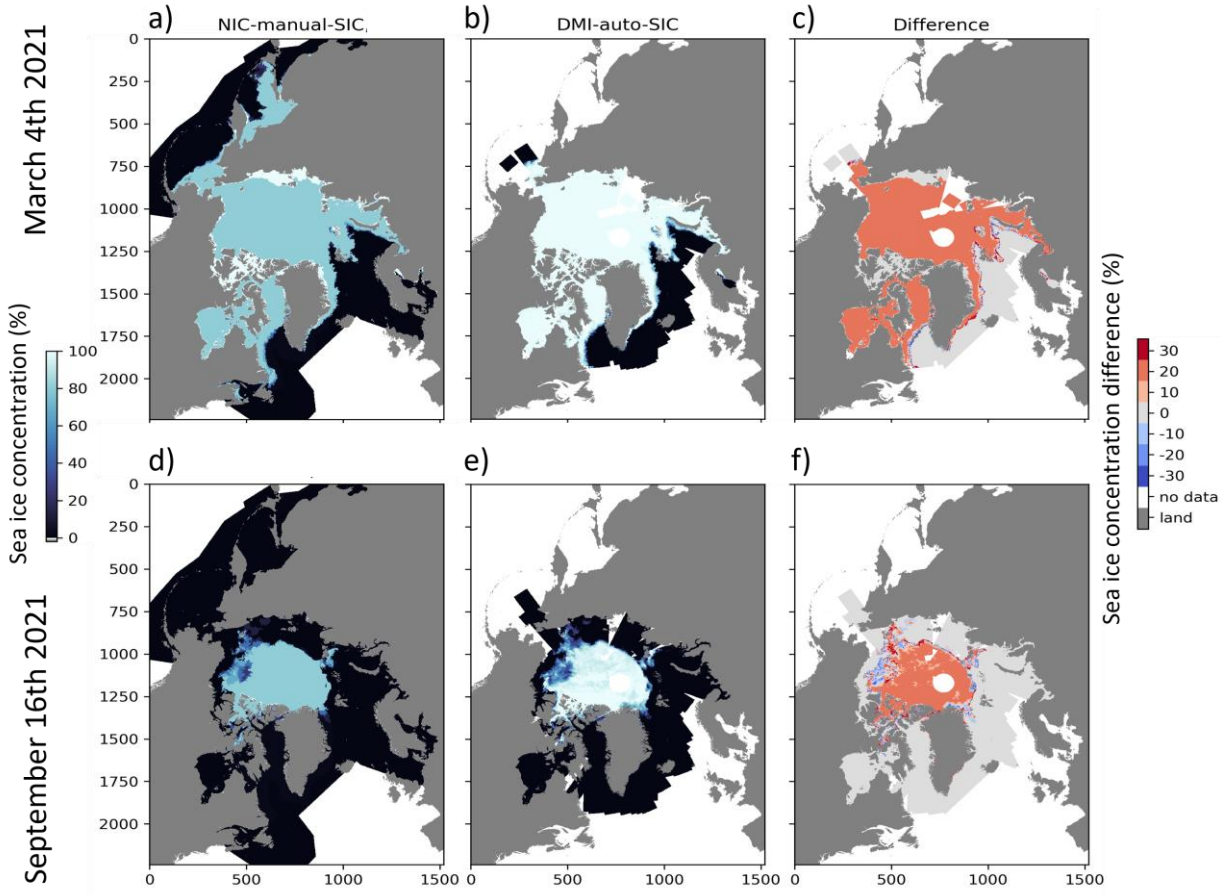

Figure 6 – Sea ice concentration comparison of NIC ice charts (a)(b) with the L3 product (b)(e) and the sea ice concentration difference (c)(f). Examples from a day in winter in (a)(b)(c) on March 4th 2021 and for a day in summer (d)(e)(f) on September 16th 2021. The scale bar of

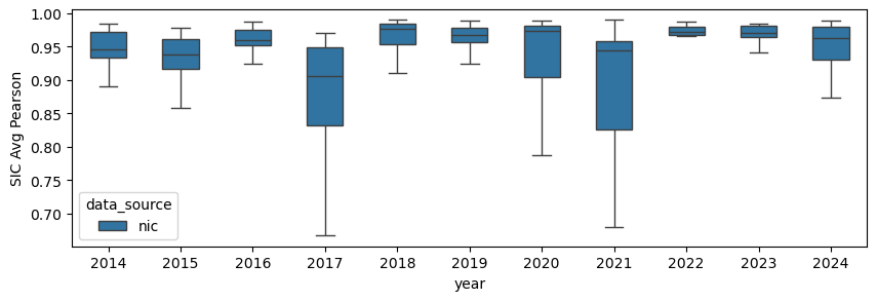

Figure 7 - Time series plot of the annual average Pearson correlation coefficient for the sea ice concentration (SIC) comparison of NIC ice charts with the L3 product, 2014 to 2024.

Table 4 - Annual average Pearson's correlation coefficient and standard deviation for the sea ice concentration (SIC) comparison of NIC ice charts with the L3 product, 2014 to 2024.

Year	Annual average Pearson's coefficient and std.dev.
Interannual average	0.93 ± 0.07
2014	0.94 ± 0.04
2015	0.93 ± 0.04
2016	0.95 ± 0.07
2017	0.86 ± 0.12
2018	0.95 ± 0.07
2019	0.96 ± 0.04
2020	0.91 ± 0.12
2021	0.87 ± 0.13
2022	0.97 ± 0.01
2023	0.97 ± 0.02
2024	0.94 ± 0.07

IV.1.2 Comparison with DMI ice charts

The L3 product SIC values are compared to the DMI ice chart SIC values for all timely matches of the two datasets throughout the period January 1st 2022 to October 3rd 2024, and the SIC linear correlation is assessed by calculating the Pearson's correlation coefficient. Figure 9 shows an overview of assessment results in terms of the annual average Pearson's coefficient and standard deviation in 2022, 2023 and 2024. The same numbers are provided in Table 5. Annual average correlation coefficients are between 0.91 and 0.93, suggesting a very high correlation, with an inter-annual average of 0.92 ±0.03.

Figure 11 (in chapter IV 1.3) highlights the seasonality of the correlation by a time series plot of the monthly average Pearson's coefficient for the sea ice concentration (SIC) comparison of DMI ice charts (and OSI SAF product and NIC ice charts) with the L3 product for one year (2021). Note that DMI ice charts from 2021 were also used in the training of the DMI-ASIP model that produces the L3 product, so this assessment is not independent. However, monthly average correlation coefficients are in the same order as those of the NIC ice chart assessment: from 0.86 to 0.97, suggesting a high correlation. As for the NIC assessment, correlations are a bit higher in summer months than in the rest of the year, but it is less pronounced for the DMI assessment. This is likely caused by the use of DMI ice charts in the training. Figure 8 shows example plots of the L3 product SIC, DMI ice chart SIC and the SIC difference, where the top panel is a day in winter (March 1st 2024) and bottom panel is a day in summer (September 1st 2024).

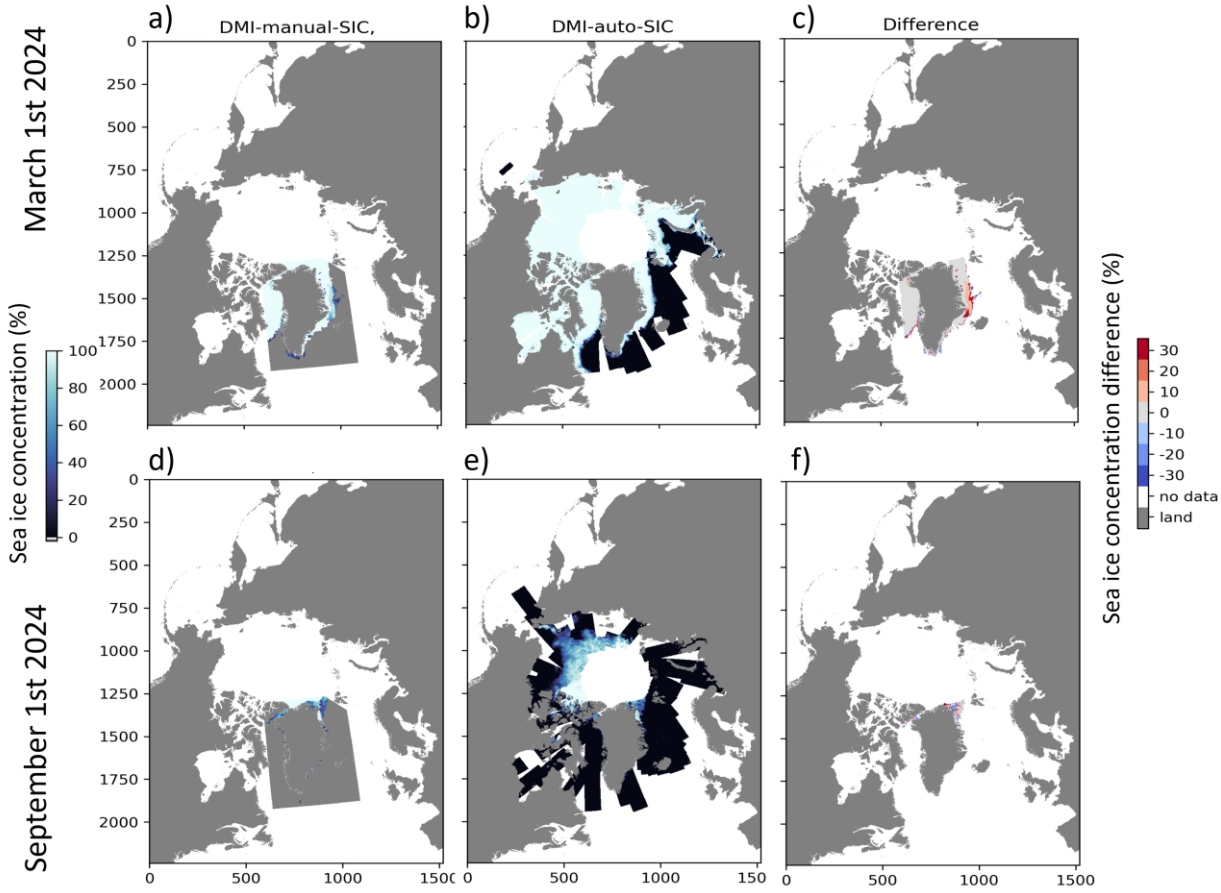

Figure 8 - Sea ice concentration comparison of DMI ice charts (a)(d) with the L3 product (b)(e) and the sea ice concentration difference (c)(f). Examples from winter in (a)(b)(c) on March 1st 2024 and for summer in (d)(e)(f) on September 1st 2024.

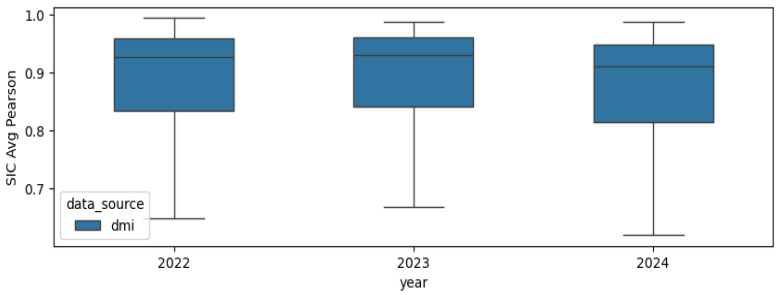

Figure 9 - Time series plot of the annual average Pearson coefficient for the sea ice concentration (SIC) comparison of DMI ice charts with the L3 product, 2022 to 2024.

Table 5 - Annual average Pearson's correlation coefficient and standard deviation for the sea ice concentration (SIC) comparison of DMI ice charts with the L3 product, 2022 to 2024.

Year	Annual average Pearson correlation coefficient and std.dev.
Interannual average	0.92 ± 0.03
2022	0.92 ± 0.03
2023	0.93 ± 0.03
2024	0.91 ± 0.04

IV.1.3 Comparison with OSI SAF product

The L3 product SIC values are compared to the OSI SAF SIC values for the daily dataset matches throughout the year 2021, and as for the NIC and DMI ice charts assessments, the SIC linear correlation is assessed by calculating the Pearson's correlation coefficient. Figure 11 shows the seasonality of the correlation through displaying a time series plot of the monthly average Pearson's coefficient and standard deviation. The same numbers are provided in Table 6. Monthly average correlation coefficients range between 0.83 and 0.89, suggesting a high correlation, with an annual average of 0.85 and a standard deviation of 0.02. The high correlation applies for all months, only the sea ice minimum-month of September and also October showing slightly lower correlation. The annual average correlation for the assessment with OSI SAF in 2021 is however in the lower range, when compared to the results from the SIC assessments against NIC and DMI ice charts. However, the assessment against NIC ice charts also had the lowest correlation in the same year of 2021. Figure 10 shows example plots of the L3 product SIC, OSI SAF SIC and the SIC difference in winter (top panel, March 1st 2024) and in summer (bottom panel, September 18th 2024). As these plots also show, the largest differences between the two datasets are in the marginal ice zone, where the L3 product systematically provide higher SIC values than the OSI SAF product. This is explained by the coarser resolution OSI SAF product not being able to resolve the ice cover at intermediate and low SICs. The correlation is even lower after the summer, in September, where the sea ice cover is thinned and experience intermediate and low SICs in the wider Arctic Ocean.

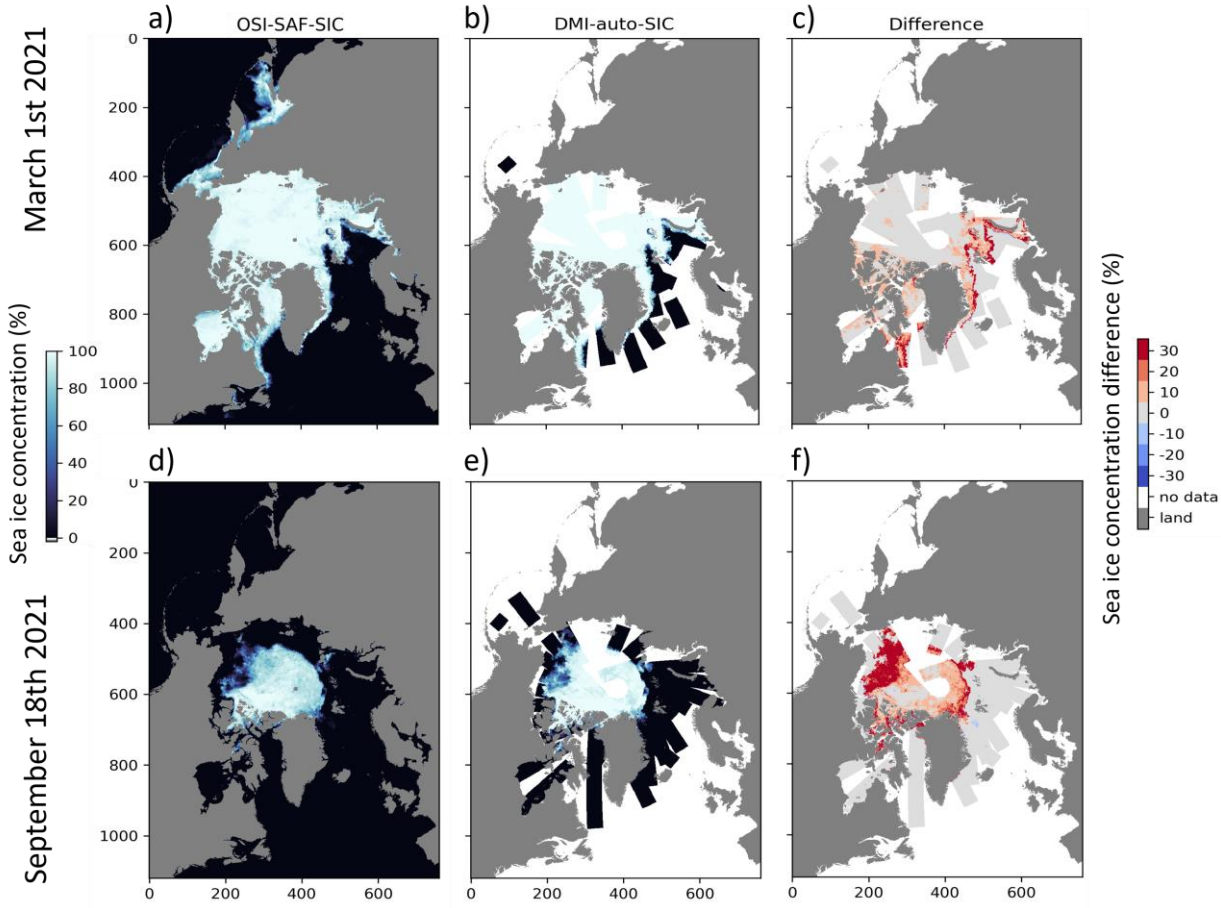

Figure 10 - Sea ice concentration comparison of OSI SAF product (a)(d) with the L3 product (b)(e) and the sea ice concentration difference (c)(f). Examples from winter in (a)(b)(c) on March 1st 2021 and for summer in (d)(e)(f) on September 18th 2021.

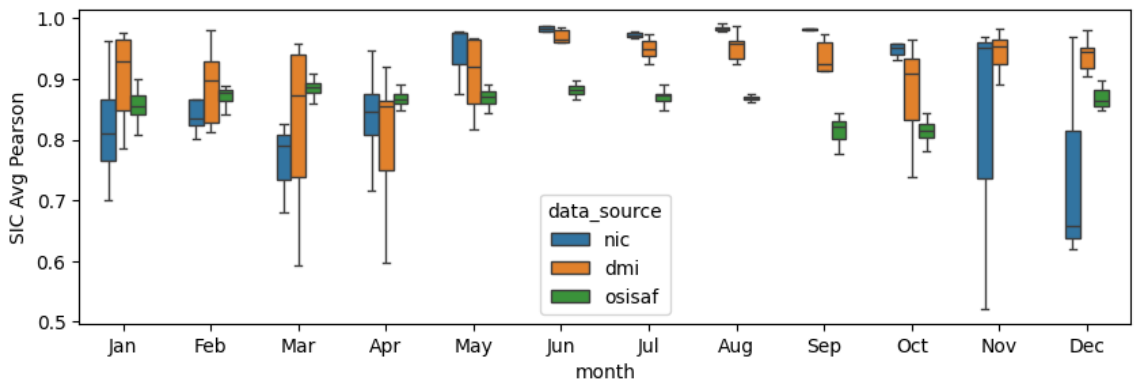

Figure 11 - Time series plot of the monthly average Pearson's correlation coefficient for the sea ice concentration (SIC) comparison of OSI SAF product (green), NIC ice charts (blue) and DMI ice charts (orange) with the L3 product in 2021.

Table 6 - Monthly average Pearson correlation coefficient and standard deviation for the sea ice concentration (SIC) comparison of the OSI SAF SIC product with the L3 product in 2021.

Month	Annual average Pearson correlation coefficient and std.dev.
2021 annual average	0.85 ± 0.02
January	0.86 ± 0.01
February	0.88 ± 0.01
March	0.89 ± 0.01
April	0.87 ± 0.01
May	0.87 ± 0.01
June	0.88 ± 0.01
July	0.87 ± 0.01
August	0.87 ± 0.01
September	0.83 ± 0.02
October	0.83 ± 0.01
November	0.85 ± 0.01
December	0.85 ± 0.02

IV.2 Sea Ice Stage-of-Development

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product sea ice stage-of-development (SoD) variable is compared to NIC ice charts in the years 2014 to 2024, and to DMI ice charts from 2022 to 2024. The assessment results in the form of SoD class difference plots and F1 score statistics are presented and discussed in the following sections.

IV.2.1 Comparison with NIC ice charts

The L3 product compared with NIC ice charts in the winter (March 4th 2021) and for the summer (September 16th 2022) are shown in Figure 12. The difference plots use a scale ranging from -3 to 3, where 3 indicates the L3 product having an older/thicker SoD class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a thinner SoD class than the ice chart. Both maps show reasonably good agreement in both summer and winter, with the L3 product having a tendency towards providing an older/thicker SoD class than the ice chart.

Figure 13 shows a confusion matrix for the L3 product SoD classes (on the y-axis) against the corresponding NIC ice chart SoD classes (on x-axis) in the period October 3rd 2014 to October 2nd 2024. The colour intensity in each square indicates the number of matches of the SoD classes. The diagonal represents the correct SoD class matches. It can be seen that the best correspondence is found for the classes thick first-year ice (FYI) and multi-year ice (MYI). There is a weaker correlation for the two thinnest SoD classes, young Ice (YI) and thin first-year ice. The lack of correlation for the glacier Ice SoD class is likely due to the category being less well represented in both datasets.

Figure 14 shows a time series plot of the annual average F1 score and standard deviation that indicate the reliability of the L3 product SoD classes. Figure 15 a) to d) shows the corresponding time series plots of annual average F1 score and standard deviation for each of the five SoD classes. The statistical values

shown in Figure 14 and Figure 15 are also provided in Table 7. The numbers show the same as seen from the confusion matrix, that the best correspondence is found for the classes thick first-year ice (FYI) and multi-year ice (MYI), with an inter-annual average F1 scores of 0.60 and 0.80. There is a weaker correlation for the two thinnest SoD classes, young Ice (YI) and thin first-year ice, and the glacier ice class. This could partly be due to a bad temporal alignment of the input data used for the daily L3 product and the weekly NIC ice chart. The difference in ice charting methodology and use of the different SoD classes, compared to the NIC ice chart dataset that was used for the training of the DMI-ASIP model, could also be part of the explanation. The weak performance for these three SoD classes results in an overall SoD inter-annual average F1 score of 0.42 ± 0.07, which indicate a poor performance.

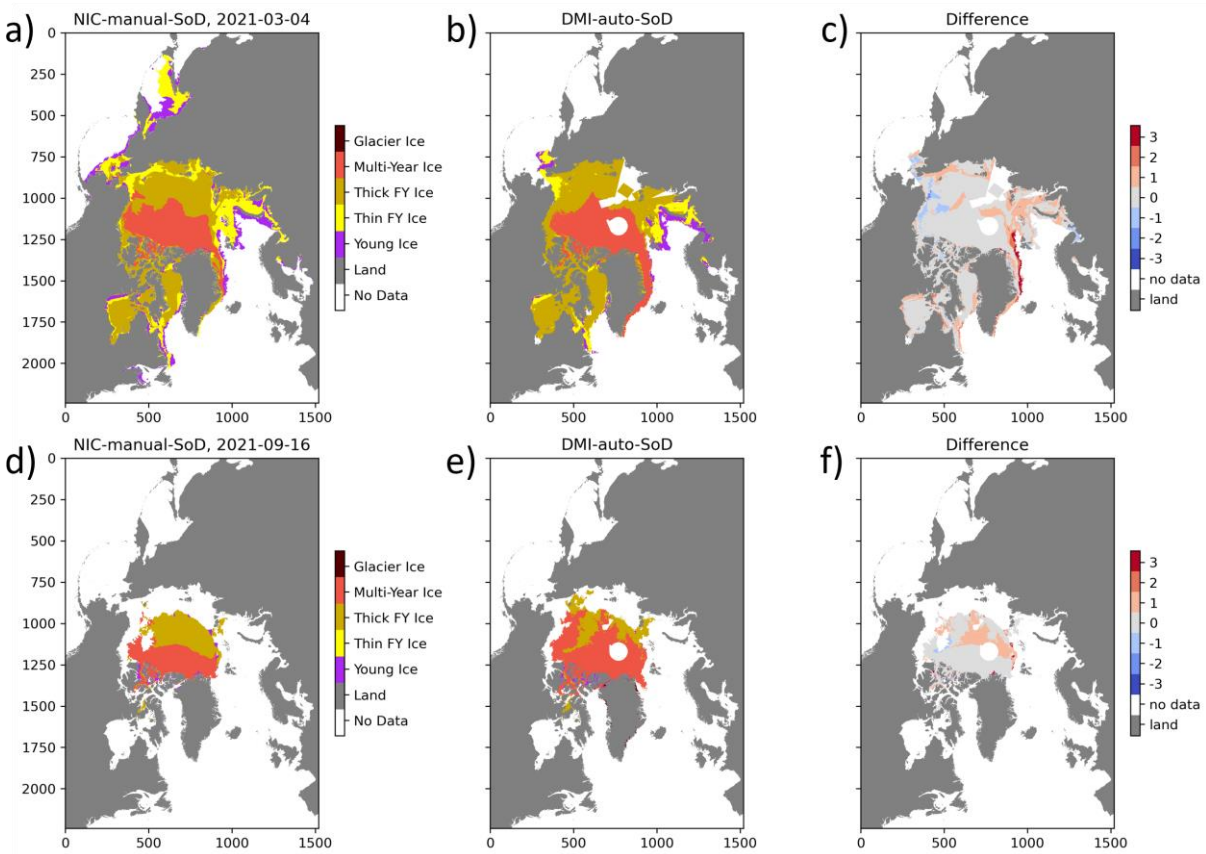

Figure 12 - Sea ice stage-development (SoD) classes comparison of NIC ice charts (a)(d)) with the L3 product (b)(e) and the stage-of-development class difference (c)(f). Examples from a day in winter (a)(b)(c) on March 4th 2021 and for a day in summer in (d)(e)(f) on September 16th 2021. The difference plots use a scale ranging from -3 to 3, where 3 indicates the L3 product having an older/thicker SoD class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a thinner SoD class than the ice chart.

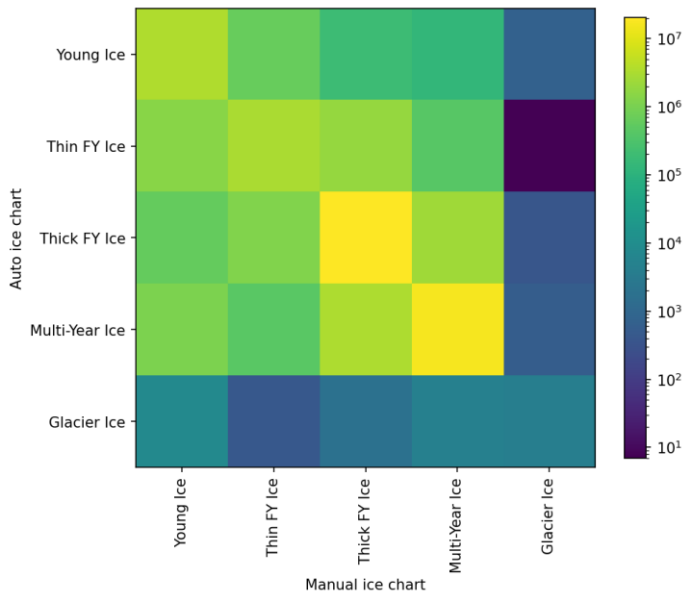

Figure 13 – Confusion matrix from the results of the sea ice stage-development (SoD) classes comparison of NIC ice charts with the L3 product. The colour intensity in each square indicates the number of matches of the SoD classes. The diagonal represents the correct SoD class matches.

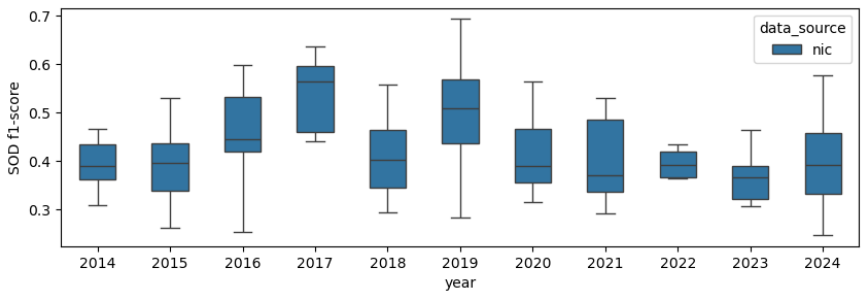

Figure 14 – Time series plot of annual average F1-scores for the sea ice stage-development (SoD) classes comparison of NIC ice charts with the L3 product, October 2014 to October 2024.

Table 7 - Annual average F1 score for the comparison between the sea ice stage-development (SoD) classes of NIC ice charts with the L3 product, period 2014 to 2024. Interannual average through 2014-2014 ± standard deviation.

F1 ±Std Year	SoD annual average	Young Ice	Thin FY ice	Thick FY Ice	Multi-Year Ice	Glacier Ice
2014	0.39 ±0.05	0.27 ±0.18	0.09 ±0.06	0.07 ±0.04	0.87 ±0.08	0.70 ±0.31
2015	0.39 ±0.07	0.45 ±0.14	0.20 ±0.20	0.58 ±0.28	0.82 ±0.05	0.30 ±0.18
2016	0.46 ±0.09	0.37 ±0.17	0.15 ±0.11	0.84 ±0.17	0.80 ±0.10	0.02 ±0.01
2017	0.54 ±0.08	0.62 ±0.18	0.49 ±0.24	0.50 ±0.40	0.82 ±0.09	0.40 ±0.29
2018	0.41 ±0.08	0.67 ±0.19	0.41 ±0.34	0.62 ±0.29	0.84 ±0.05	0.07 ±0.05
2019	0.50 ±0.11	0.46 ±0.18	0.39 ±0.22	0.74 ±0.18	0.81 ±0.05	0.62 ±0.37
2020	0.42 ±0.08	0.63 ±0.08	0.48 ±0.24	0.54 ±0.26	0.87 ±0.03	0.18 ±0.14
2021	0.40 ±0.09	0.53 ±0.15	0.30 ±0.16	0.47 ±0.39	0.84 ±0.04	0.17 ±0.00
2022	0.40 ±0.03	0.12 ±0.11	0.01 ±0.01	0.90 ±0.03	0.74 ±0.03	-
2023	0.36 ±0.04	0.57 ±0.28	0.22 ±0.12	0.65 ±0.30	0.75 ±0.08	0.75 ±0.30
2024	0.39 ±0.09	0.42 ±0.19	0.25 ±0.14	0.73 ±0.22	0.60 ±0.07	0.68 ±0.48
Interannual average	0.42 ±0.07	0.46 ±0.17	0.27 ±0.17	0.60 ±0.23	0.80 ±0.06	0.39 ±0.21

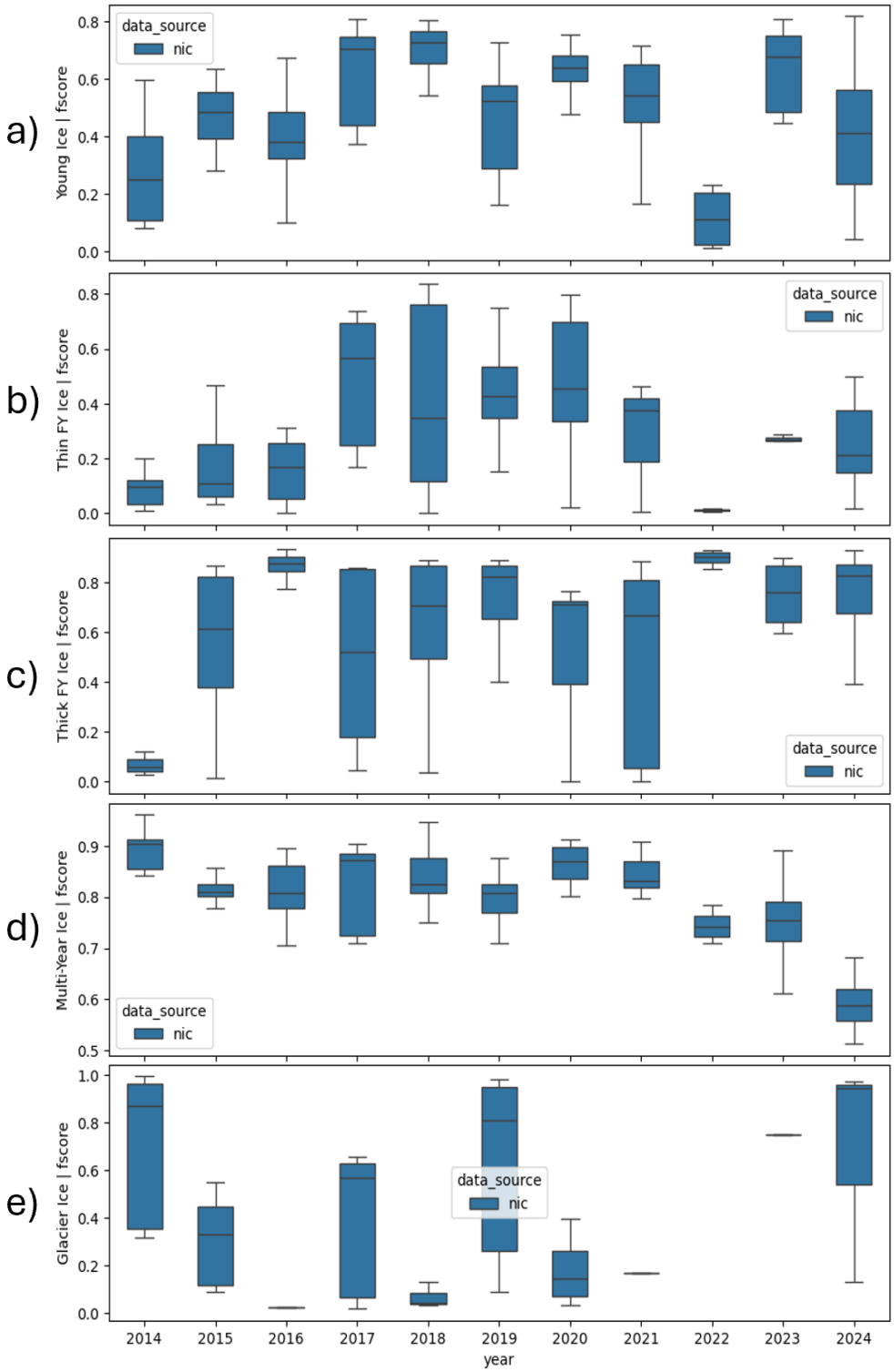

Figure 15 – Time series plot of annual average F1-scores for the five sea ice stage-of-development (SoD) classes comparison of DMI ice charts with the L3 product, January 2022 to October 2024 : a) young ice, b) thin first-year ice, c) thick first-year ice, d) multiyear ice and e) glacier ice.

IV.2.2 Comparison with DMI ice charts

Comparisons between the L3 product and the DMI ice charts, both in the winter (March 1st 2024) and in the summer (September 1st 2024) are shown in **¡Error! No se encuentra el origen de la referencia.**. The difference plots use a scale ranging from -3 to 3, where 3 indicates the L3 product having an older/thicker SoD class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a thinner SoD class than the ice chart. Both maps show reasonably good agreement in winter and summer (for the small sea ice area that is covered), with the L3 product having a tendency towards providing an older/thicker SoD class than the ice chart.

Figure 17 shows a confusion matrix for the L3 product SoD classes (on the y-axis) against the corresponding DMI ice chart SoD classes (on x-axis) in the period January 1st 2022 to October 2nd 2024. The colour intensity in each square indicates the number of matches of the SoD classes. The diagonal represents the correct SoD class matches. As with the NIC ice chart assessment, the best correspondence is found for the classes thick first-year ice (FYI) and multi-year ice (MYI). There is a weaker correlation for the two thinnest SoD classes, young Ice (YI) and thin first-year ice. The lack of correlation for the glacier Ice SoD class is likely due to the category being less good represented in both of the datasets.

Figure 18 shows a time series plot of the annual average F1 score and standard deviation that indicate the reliability of the L3 product SoD classes. Figure 19 a) to d) show the corresponding time series plots of annual average F1 score and standard deviation for each of the five SoD classes. The statistical values shown in Figure 18 and Figure 19 are also provided in Table 8. In agreement with the confusion matrix, the best correspondence is found for the classes thick first-year ice (FYI) and multi-year ice (MYI), with an inter-annual average F1 scores of 0.61 and 0.81. There is a weaker correlation for the two thinnest SoD classes, young Ice (YI) and thin first-year ice, and the glacier ice class. The YI and thin FYI classes are only represented in the datasets during certain part of the year, and the glacier ice class is only represented in minor areas along the Greenland coast. Thus, the reason for the poor performance for these three classes could be explained by their underrepresentation, compared to the thick FYI and MYI classes. The weak performance for these three SoD classes results in an overall SoD inter-annual average F1 score of 0.45 ±0.10, which indicate a poor performance. These results are consistent with the results of the SoD assessment against NIC ice charts.

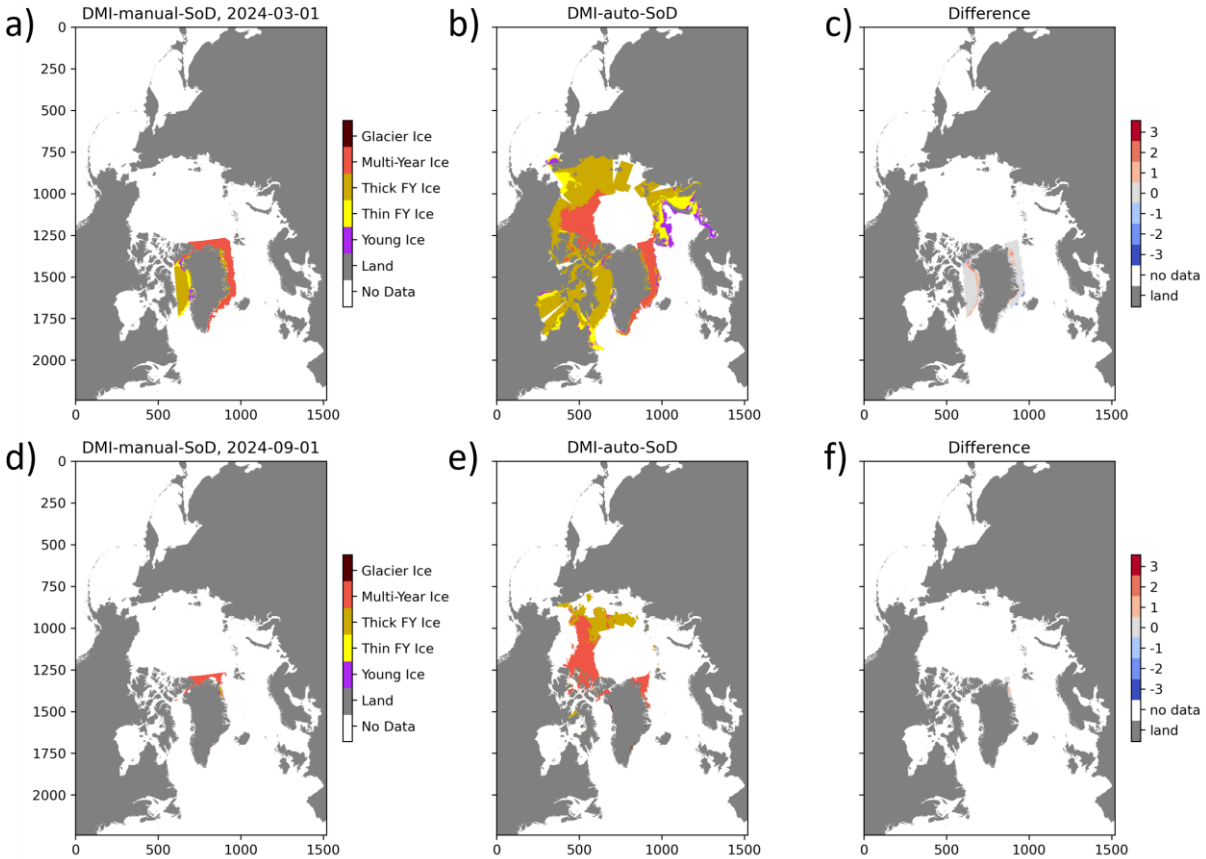

Figure 16 - Sea ice stage-development (SoD) classes comparison of DMI ice charts (a)(d) with the L3 product (b)(e) and the stage-of-development class difference (c)(f). Examples from a day in winter in (a)(b)(c) on March 1st 2024 and for a day in summer in (d)(e)(f) on September 1st 2024. Scale ranging from -3 to 3, where 3 indicates the L3 product having an older/thicker SoD class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a thinner SoD class than the ice chart.

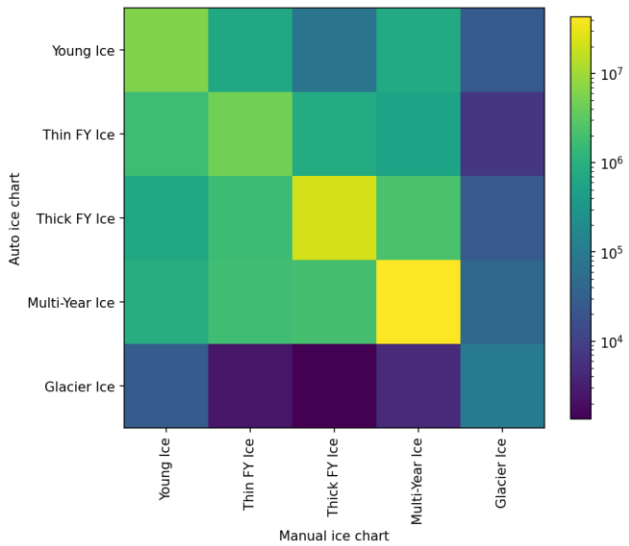

Figure 17 – Confusion matrix from the results of the sea ice stage-development (SoD) classes comparison of DMI ice charts with the L3 product. The colour intensity in each square indicates the number of matches of the SoD classes. The diagonal represents the correct SoD class matches.

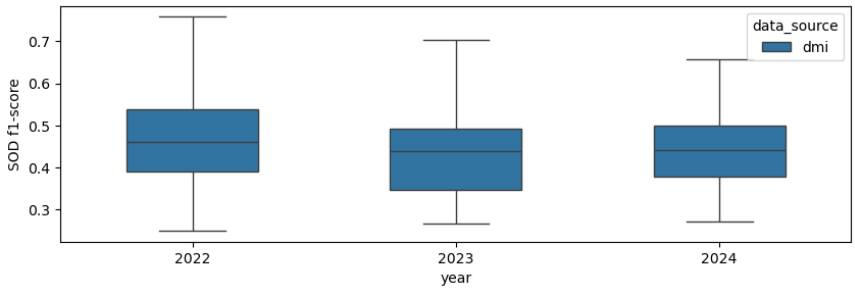

Figure 18 – Time series plot of annual average F1-scores for the sea ice stage-development (SoD) classes comparison of DMI ice charts with the L3 product, October 2014 to October 2024.

Table 8 - Annual average F1 score for the sea ice stage-development (SoD) classes comparison of DMI ice charts with the L3 product, 2022 to 2024. Interannual average through 2014-2024 ± standard deviation.

F1 ±Std Year	SoD annual average	Young Ice	Thin FY ice	Thick FY Ice	Multi-Year Ice	Glacier Ice
2022	0.48 ±0.12	0.43 ±0.26	0.34 ±0.25	0.61 ±0.31	0.80 ±0.11	0.47 ±0.35
2023	0.43 ±0.09	0.40 ±0.25	0.42 ±0.24	0.58 ±0.30	0.79 ±0.14	0.43 ±0.33
2024	0.45 ±0.09	0.34 ±0.15	0.29 ±0.24	0.64 ±0.30	0.84 ±0.10	0.40 ±0.27
Interannual avg.	0.45 ±0.10	0.39 ±0.22	0.32 ±0.24	0.61 ±0.30	0.81 ±0.12	0.43 ±0.31

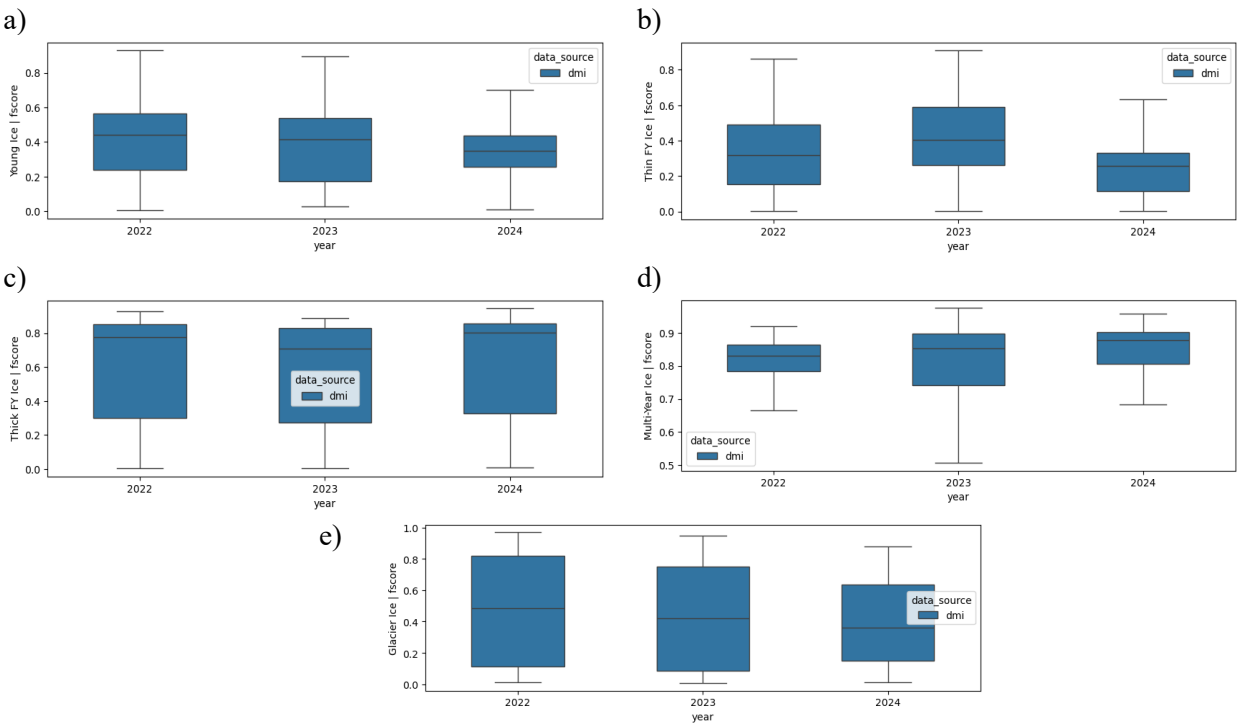

Figure 19 – Time series plot of annual average F1-scores for the five sea ice stage-of-development (SoD) classes comparison of DMI ice charts with the L3 product, January 2022 to October 2024: a) young ice, b) thin First-year ice, c) thick first-year ice, d) multiyear ice and e) glacier ice.

IV.3 Sea Ice Floe Size

The daily *Arctic Ocean - High Resolution Sea Ice Information L3* product sea ice floe size (Flz) variable is compared to DMI ice charts in the years 2022 to 2024. Few ice chart datasets contain the Flz information, the DMI ice charts being one of those datasets. The assessment results in the form of Flz class difference plots and F1 score statistics are presented and discussed in the following section. The assessment of comparing the products four Flz classes against the DMI ice chart dataset show best correspondence for the Flz class vast & giant floes, that is widely used in the L3 product and by the ice chart for the pack ice region in Central Arctic, Northeast and Northwest Greenland ice areas. Despite the good and reliable performance for the vast & giant floe class, the weak performance of three smaller Flz classes results in an overall inter-annual average F1 score of 0.39 ± 0.08, which indicate a poor performance.

IV.3.1 Comparison with DMI ice charts

The L3 product is compared with DMI ice charts in the winter (March 1st 2024) and in the summer (September 1st 2024) in . The difference plots use a scale ranging from -3 to 3, where 3 indicates the L3 product having a larger Flz class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a smaller Flz class than the ice chart.

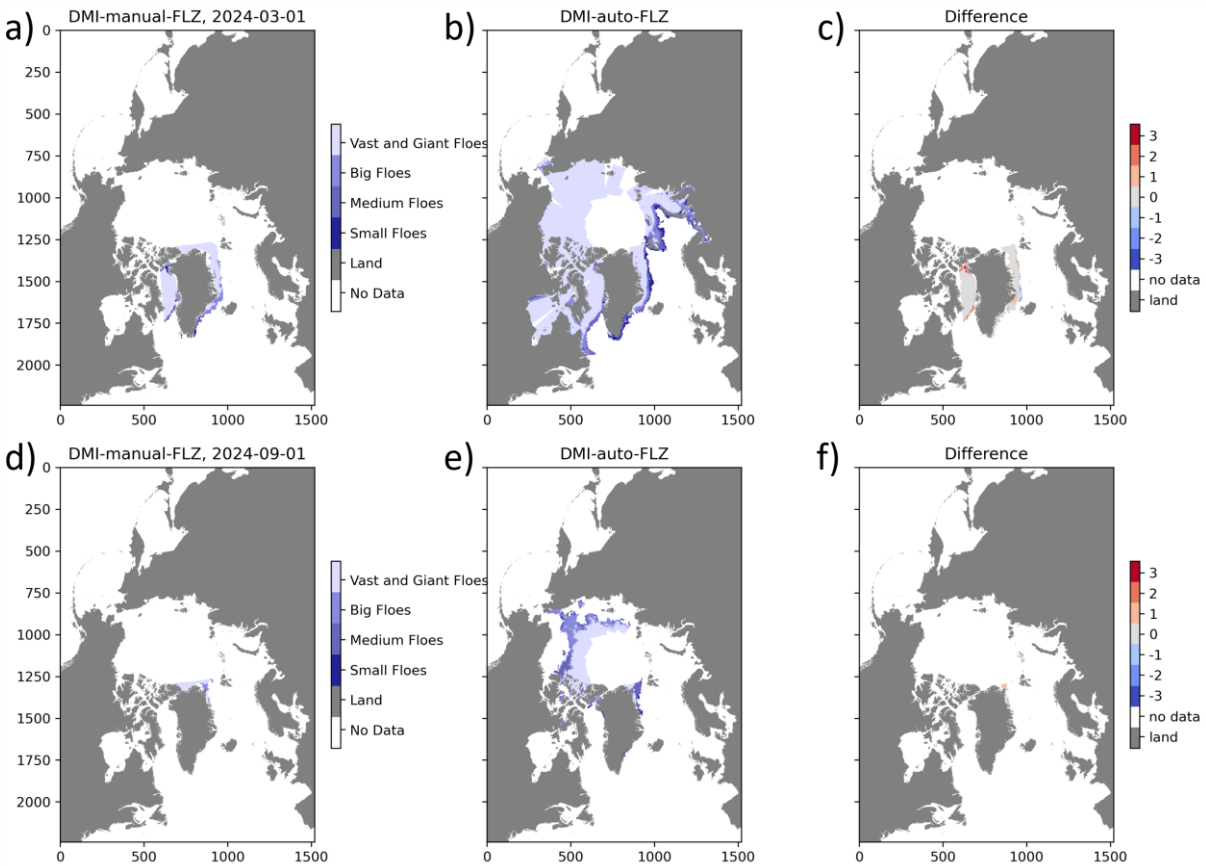

Figure 20 - Sea ice floe size (Flz) classes comparison of DMI ice charts (a)(d) with the L3 product (b)(e) and the stage-of-development class difference (c)(f). Examples from a day in winter in (a)(b)(c) on March 1st 2024 and for a day in summer in (d)(e)(f) on September 1st 2024. The difference plots use a scale ranging from -3 to 3, where 3 indicates the L3 product having a larger Flz class than the ice chart, 0 represents perfect agreement, and -3 signifies the L3 product having a smaller Flz class than the ice chart.

Both maps show reasonably good agreement in winter and summer across the small overlapping sea ice area that is covered, with the L3 product having a tendency towards providing a larger Flz class than the ice chart.

Figure 21 shows a confusion matrix for the L3 product Flz classes (on the y-axis) against the corresponding DMI ice chart Flz classes (on x-axis) in the period January 1st 2022 to October 2nd 2024. The colour intensity in each square indicates the number of matches of the Flz classes. The diagonal represents Flz class agreement between the two datasets. The best correspondence is found for the vast & giant floe class, that is widely used in the L3 product and by the ice chart for pack ice region in Central Arctic, Northeast and Northwest Greenland ice areas. There is a weaker correlation for the three smaller Flz classes, at least when compared to the well-represented vast & giant floe class.

Figure 22 shows a time series plot of the annual average F1 score and standard deviation that indicate the reliability of the L3 product Flz classes. Figure 23 a) to d) shows the corresponding time series plots of annual average F1 score and standard deviation for each of the four Flz classes. The statistical values shown in Figure 22 and Figure 23 are also provided in Table 9. In agreement with the confusion matrix, the best correspondence is found for the vast & giant floe class with an inter-annual average F1 score of 0.87. There is a weak correlation for the three smaller Flz classes, with F1 scores in the order of 0.13 to 0.30. Since those three smaller floe size categories are primarily found in the more dynamical marginal ice zone, any time difference between the ice chart and L3 product can result in Flz class mismatch on the grid level and this may to some degree explain the low performance for those three classes. Despite the good and reliable performance for the vast & giant floe class, the weak performance of the three smaller Flz classes results in an overall inter-annual average F1 score of 0.39 ± 0.08, which indicate a poor performance.

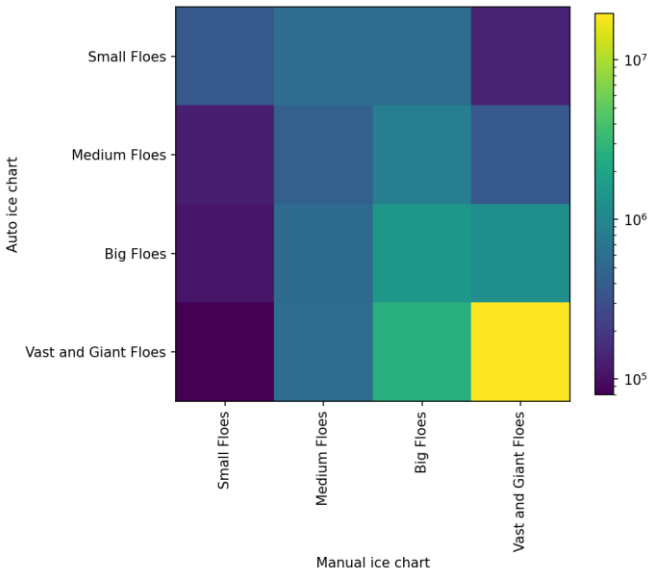

Figure 21 – Confusion matrix from the results of the sea ice floe size (Flz) classes comparison of DMI ice charts with the L3 product, January 2022 to October 2024. The colour intensity in each square indicates the number of matches of the Flz classes. The diagonal represents Flz class agreement between the two datasets.

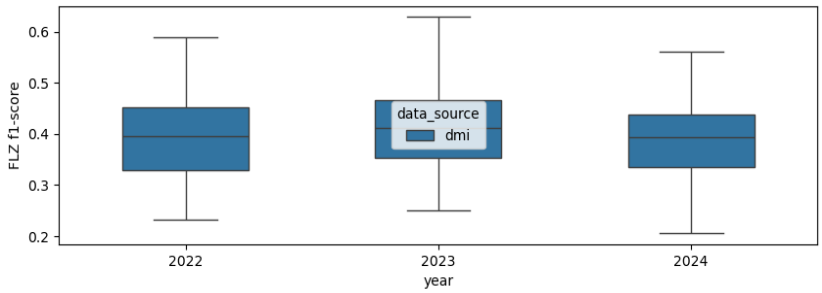

Figure 22 – Time series plot of annual average F1-scores for the sea ice floe size (Flz) classes comparison of DMI ice charts with the L3 product, January 2022 to October 2024.

Table 9 - Annual average F1 score for the sea ice floe size (Flz) classes comparison of DMI ice charts with the L3 product, 2022 to 2024. Interannual average through 2014-2024 ± standard deviation.

F1 ±Std.dev. Year	SOD annual average	Small floes	Medium floes	Big floes	Vast & Giant floes
2022	0.39 ±0.08	0.26 ±0.17	0.20 ±0.14	0.30 ±0.15	0.88 ±0.06
2023	0.39 ±0.08	0.33 ±0.21	0.19 ±0.13	0.29 ±0.15	0.89 ±0.05
2024	0.39 ±0.07	0.33 ±0.19	0.13 ±0.10	0.28 ±0.12	0.85 ±0.08
Interannual avg.	0.39 ±0.08	0.29 ±0.19	0.17 ±0.13	0.29 ±0.14	0.87 ±0.06

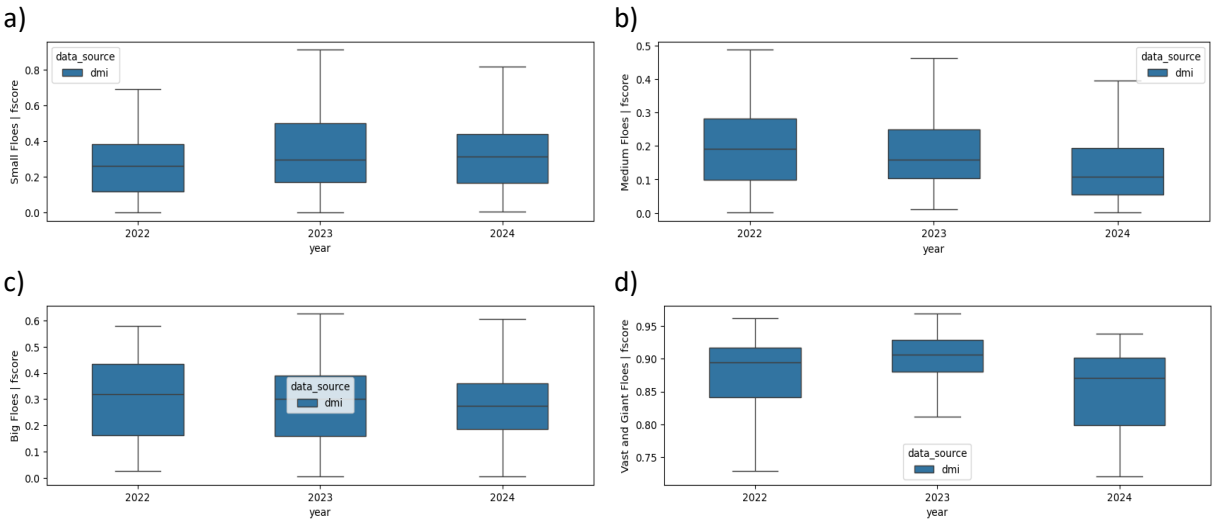

Figure 23 – Time series plot of annual average F1-scores for the sea ice floe size (Flz) classes comparison of DMI ice charts with the L3 product, January 2022 to October 2024: a) small floes, b) medium floes, c) big floes and d) vast and giant floes.

V BIAS STUDY: SENTINEL-1 VS. RADARSAT CONSTELLATION MISSION

For the ingestion of new upstream data from the Radarsat Constellation Mission (RCM) SAR imagery with entry-into-service in November 2025, a comparative analysis was conducted between DMI-ASIP deep learning model outputs derived from Sentinel-1 (S1) and RCM, respectively, to assess their consistency and potential systematic differences, when introducing the use of RCM SAR imagery, in addition to Sentinel-1, as input to the DMI-ASIP model for the EiS in November 2025. The comparison of SIC outputs shows a very high degree of correlation with no significant systematic under- or over-estimation of SIC by one sensor relative to the other. Summarizing statistics for the SoD and FLZ variables demonstrate moderate to strong agreement and comparable classification performances.

V.1 Dataset and Methodology

The evaluation was carried out using a dataset of 443 pairs of coincident S1 and RCM acquisitions (see Figure 24) from 2020 through 2023. These pairs were acquired within a 5-minute timeframe of each other to ensure comparable sea ice conditions. The S1 and RCM images were ingested in the DMI-ASIP model to produce L2 outputs on an 80-meter grid. The L2 outputs were compared at the level of a grid cell by resampling the L2 output based on RCM to grid of the L2 output based on S1. The cross-sensor consistency was quantified using the coefficient of determination (R^2, the square of the Pearson correlation coefficient, R), mean bias (difference) and the Root Mean Squared Difference (RMSD, the square root of the mean squared difference, measuring the standard deviation of the difference) for the sea ice concentration (SIC) parameter, and confusion matrices and the Mean Intersection over Union (mIoU) for the categorical parameters, sea ice stage-of-development (SoD) and sea ice floe size. IoU is calculated by dividing the overlapping area (grid cells) between the product and reference data classes by the total area covered by both. MIoU is a multi-class metric, that takes the IoU value for each class in the product and calculates the average to provide an overall performance score.

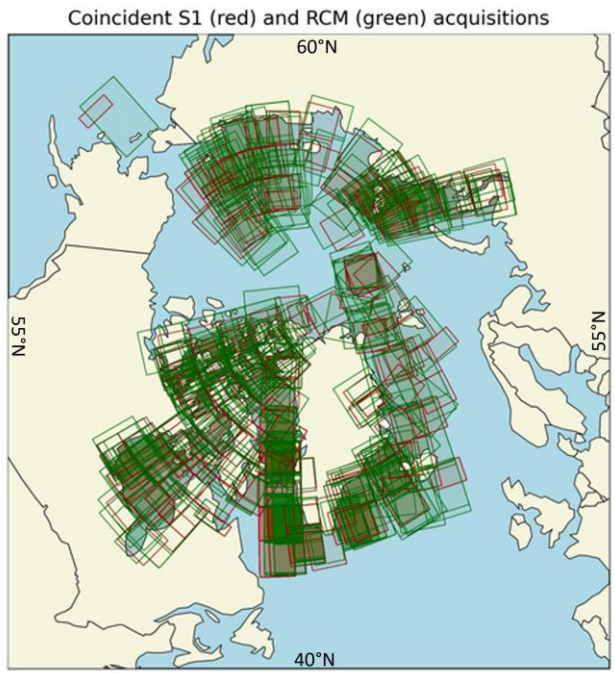

Figure 24 - Geographical distribution in the Arctic of the coincident pairs of Sentinel-1 imagery (shown as green bounding boxes) and Radarsat Constellation Mission imagery (red bounding boxes) acquisitions used in the bias study.

V.2 Results and Analysis

V.2.1 Sea Ice Concentration

The comparison of SIC outputs shows a very high degree of correlation with no significant systematic under- or over-estimation of SIC by one sensor relative to the other.

- An R^2 of 99.9% was found, indicating a near-perfect linear relationship between the two sensors.
- The RMSD was calculated to be 1.48% with a mean bias of –0.08% when comparing S1 to RCM.

Figure 25 shows the SIC bias distributions between S1 and RCM across different SIC intervals using a letter-value plot.

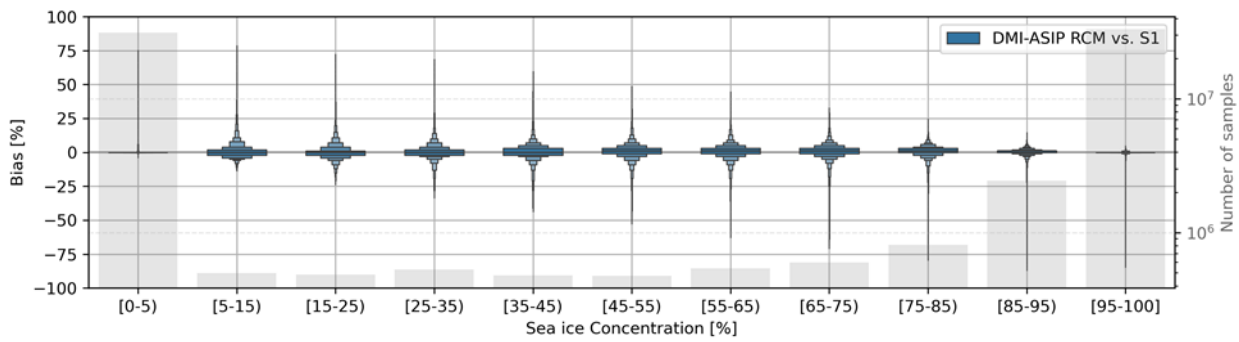

Figure 25 - Letter-value plot showing the bias distributions between Sentinel-1 and the Radarsat Constellation Mission across different sea ice concentration (as %) intervals.

V.2.2 Sea Ice Stage-of-Development and Floe Size

The results of the comparison of the sea ice stage-of-development and floe size variables are shown in confusion matrices, see Figure 26, and summarizing statistics demonstrate moderate to strong agreement and comparable classification performances.

- The mIoU was calculated to be 89.0% for the SoD and 63.2% for floe size, indicating strong and moderate levels of agreement between the sensors, respectively.

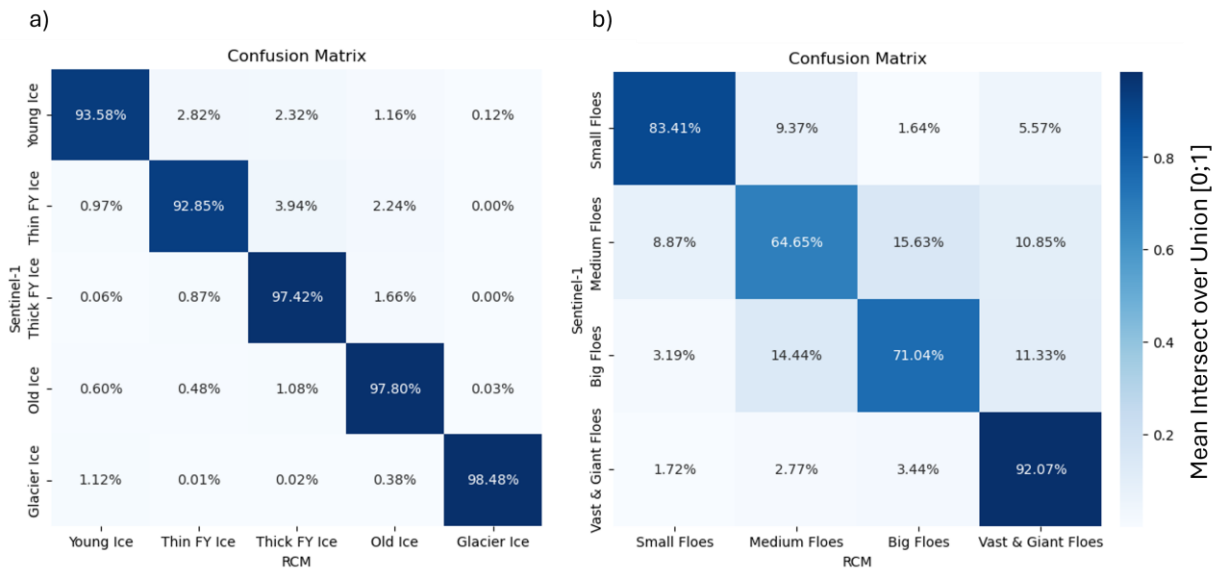

Figure 26 - Confusion matrices for (a) the Sea Ice Stage-of-Development and (b) the Sea Ice Floe Size.

VI SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to track and describe changes that may occur in the current operational system, due to outages, upstream data changes (e.g., addition or loss of a satellite, inclusion of other in-situ data).

The products covered by this QUID were initially put in production during November 2024 (version 1.0) using Sentinel-1 SAR imagery as input data. In November 2025 a slightly updated version of the system was put in operation that also use RCM SAR imagery as input.

The products´ corresponding Product User Manual [PUM, 2024] contains a list of dates where dataset files were not processed and delivered, for the L3 and L4 products MY datasets: SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 / cmems_obs-si_arc_phy_my_l3_P1D and SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 / mems_obs-si_arc_phy_my_l4_P1D.

Date	Change/Event description	System version	Other
November 2024	First version in CMEMS. Products based on Sentinel-1 SAR imagery.	V 1.0	DMI-ASIP model V 2.0
November 2025	Products based on Sentinel-1 AND Radarsat Constellation Mission SAR imagery.	V 1.1	DMI-ASIP model V 2.1

VII QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The products covered by this QUID were put in production during November 2024. There has been no change in the operational production chain (version 1.0) so far.

VIII REFERENCES

(PUM, 2024) Product User Manual for Arctic Ocean - High Resolution Sea Ice Information L3 and L4, SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023, SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024, v1.0, Copernicus Marine Service, 2024, https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-PUM-011-023-024.pdf

(QUID, 2023) Quality Information Document for the Sea Ice Thematic Assembly Center SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002, SEAICE_ANT_PHY_AUTO_L3_NRT_011_012, SEAICE_ARC_AUTO_L4_NRT_011_015, v.1.0, Copernicus Marine Service, 2023, https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-002-012.pdf

(WMO, 2010) JCOMM Technical Report No. 23: SIGRID-3: a vector archive format for sea ice charts, WMO/TD-No. 1214, 2010, https://library.wmo.int/idurl/4/37171